

Ringel's Conjecture

Walid Elkersh

July 17, 2026

Chapter 1

Statement

Definition 1.1. Ringel's conjecture for all n . (Open problem)

Definition 1.2. Ringel's conjecture for sufficiently large n .

Chapter 2

Combinatorial Primitives (Primitives.lean)

Definition 2.1. Near-difference rainbow colouring of K_{2n+1} .

Theorem 2.2. *ND-colouring is a 2-factorization.*

Lemma 2.3. $s(c+1)$ step realizes colour c .

Definition 2.4. Case predicates.

Chapter 3

Tree Structure (TreeStructure.lean)

Theorem 3.1. *A tree on n vertices has $\lceil n/(4k) \rceil$ leaves or $\lceil n/(4k) \rceil$ vertex-disjoint bare paths of length k .*

Theorem 3.2. *Connects split to case A/B/C division (MPS Lemma 2.2).*

Chapter 4

Case Division (CaseDivision.lean)

Theorem 4.1. *Routes a tree into Case A, B, or C.*

Chapter 5

Case A — Many Leaves (CaseA.lean)

Definition 5.1. Signed walk sum.

Lemma 5.2. *Tree embedding via signed edge-increment.*

Lemma 5.3. *Littlewood–Offord bound for vertex collisions.*

Lemma 5.4. *Absorption matching probability.*

Theorem 5.5. *Main Case A result.*

Chapter 6

Case B — Many Bare Paths (CaseB.lean)

Lemma 6.1. *Bare-path embedding.*

Theorem 6.2. *Main Case B result.*

Chapter 7

Case C — Small Core (CaseC.lean)

Lemma 7.1. *Greedy core embedding.*

Lemma 7.2. *Leaf extension.*

Theorem 7.3. *Main Case C result.*

Chapter 8

Probability Infrastructure (ProbBounds.lean)

Definition 8.1. Counting measure on finite types.

Lemma 8.2. *Positive probability implies existence.*

Chapter 9

Proof Assembly (Spine.lean + Proof.lean)

Theorem 9.1. *Kotzig cyclic-shift decomposition.*

Theorem 9.2. *Rainbow copy exists.*

Theorem 9.3. *Full assembly for large n .*